writer's forethought about pitfalls for the designer. You might summarize a similar project. You might suggest a design approach.

9.2 WRITING DESIGN DOCUMENTS

Once you have created a requirements specification document, you will have to decide whether to write a design document. Often it is not necessary, especially when a few people are working on a small project. You may wish to choose key elements of a complete design document and develop them to the point of usefulness.

The first part of design is concerned with the architecture of the system. The system should be broken into sections that encompass broad groups of functionality. A **Web application** for project management might break down into a module that handles project information, a module that handles users, and a module that handles timesheet entries. An informational Web site can be broken down by the secondary pages—that is, the pages one click away from the home page. The "About Us" section serves to inform visitors about the company itself, while a catalog area is a resource for learning about the items the company sells Depending on the type of site, you should choose some sort of diagram that shows the subsystems and how they relate to each other. These are called entity relationship diagrams. We almost always create a page-flow diagram. Each node in the graph is a page as experienced by the user. Lines representing links connect the page to other pages on the site. Another useful diagram is one that shows the relationship between database tables. Nodes represent tables, and you may wish to list the fields inside boxes that stand for the tables. Lines connect tables and show how fields match. It is also helpful to indicate whether the relationship between the tables is one to one or one to many.

Keyword

Web application is application software that runs on a web server, unlike computer-based software programs that are run locally on the operating system of the device

The next phase of design is interface specification. This defines how subsystems communicate. It can be as simple as listing the URLs for each page. If the site has forms, all the fields should be enumerated. If you are tracking user sessions, you will want to specify how you will be doing this, with cookies or form variables. Define acceptable values for the